

EduGuard AI

Student Dropout Predictor & Personalised Intervention Engine



HIGH RISK



MEDIUM RISK



LOW RISK



Powered by IBM Bob · watsonx.ai Foundation Model · Problem Statement E1



Team EduGuard



CHARUSAT



September 2026

Predicts · Prioritises · Intervenes · Tracks — Zero backend required · Works with any university ERP

India's Silent Crisis — College Dropout

40%

Engineering students
drop out before graduation

150+

At-risk students per
semester at CHARUSAT

₹3L

Lost per dropout in fees
and admin costs

2–3×

Semesters late when
risk is finally spotted

⚡ Root Cause

Coordinators have ALL the data — attendance, marks, financials, backlogs — but NO AI system to connect the dots and flag at-risk students before it's too late.

✗ Manual Checks

Reactive — intervenes after dropout, never before

✗ Single-Metric Alerts

Attendance only — misses financial stress & backlogs

✗ ERP Dashboards

Raw data, no AI, no risk scoring, no priority ranking

EduGuard AI — What It Does



Predicts Risk

0–100 risk score per student using 8 academic & behavioural signals — IBM Bob Agent Mode



Explains Why

Shows the exact factors driving each score — no black box, full transparency to coordinator



Ranks Students

Instantly surfaces highest-risk students so coordinators know exactly where to act first



Intervenes

IBM Bob Plan Mode generates a personalised 5-step action plan with responsible person + timeline



IBM Bob Risk Score Formula

Attendance %	× 0.28
Internal Marks	× 0.25
Assignments	× 0.15
Lab Attendance	× 0.10
Financial Stress	× 1.00
Backlogs (capped)	× 1.80
Trend Modifier	± 4 pts

Risk Level Thresholds

 HIGH RISK	66–100
Immediate critical action	
 MEDIUM RISK	36–65
Close monitoring + support	
 LOW RISK	0–35
Routine check-in	

How IBM Bob Powers EduGuard

IBM BOB AI

Genuinely
Integrated

AGENT

Agent Mode

Bob autonomously analyses 8+ student signals and builds complete risk profiles without manual queries

PLAN

Plan Mode

Before every intervention, Bob designs the optimal multi-stakeholder action sequence with priorities and deadlines

/ CMD

Slash Commands

/review [rollNo] — instant risk report · /report — weekly department digest auto-sent to all coordinators

AGENT

Subagents

Bob spawns lightweight parallel subagents for batch analysis of 500+ students — chat stays fully responsive

TASK

Background Tasks

Semester trend analysis runs in background — weekly digest reports delivered to coordinators every Monday

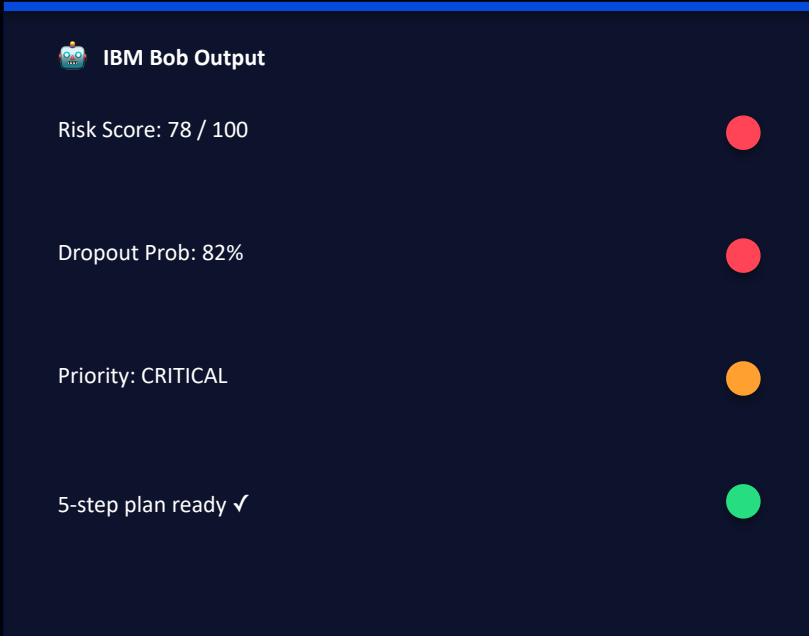
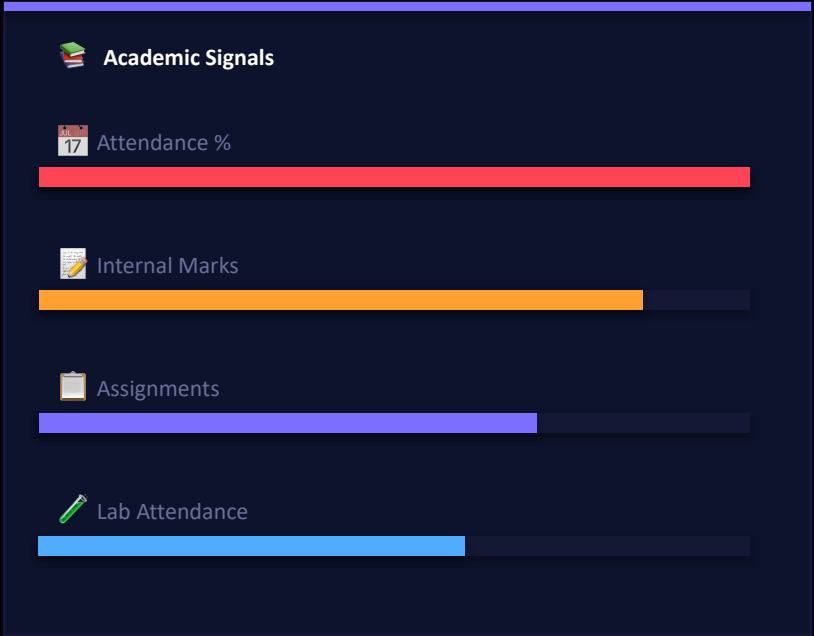
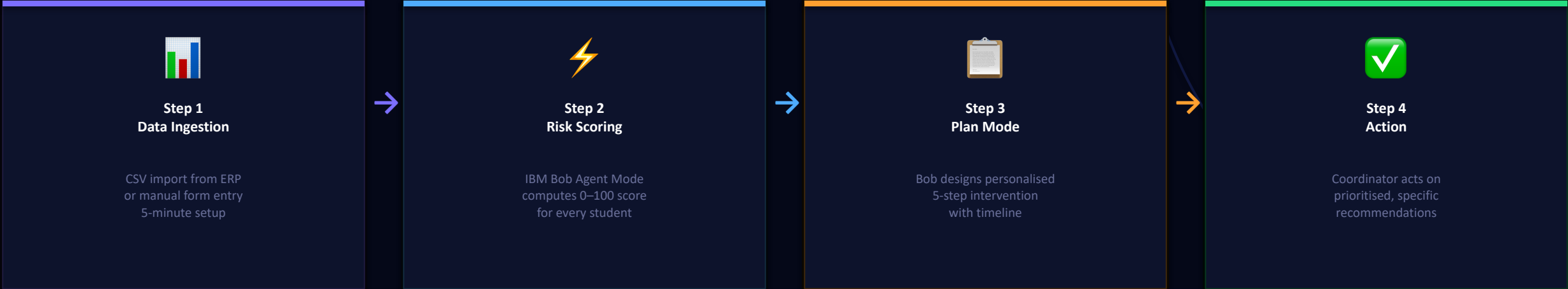
MCP

MCP Integrations

Connects to CHARUSAT FMIS (attendance), Exam Portal (marks), and Student Welfare DB via custom MCP tools

- ✓ Agent Mode
- ✓ Plan Mode
- ✓ Slash Cmds
- ✓ Subagents
- ✓ BG Tasks
- ✓ MCP

End-to-End Workflow



Built on IBM Bob & Modern Web

Layer	Technology	Role in EduGuard
AI Engine	IBM BOB	Agent · Plan · Subagents · MCP · Background Tasks
Foundation Model	watsonx.ai	NLP risk summarisation · Dropout probability estimation
Frontend	HTML5 · CSS3 · JS ES6+	Single-Page App · 4 views · Zero backend · No npm required
Charts	Chart.js 4.4	Donut · Bar · Radar · Stacked Bar — fully interactive
Data Layer	CSV Import + localStorage	Works with any ERP export · 5-minute onboarding
Design Language	Glassmorphism Dark UI	Animated SVG risk gauge · Micro-animations · Premium UX
4		0
6		∞

4 Powerful App Views



Dashboard

Real-time risk overview · High-risk student table · IBM Bob insights panel · Interactive donut chart · 4 key stats



All Students

Search & filter by risk / dept / year · CSV drag-drop import · Add student form · Export intervention reports



IBM Bob AI Chat

Conversational AI · Analyse any student · Generate intervention plans · Query department-wide risk stats



Analytics

Department comparison · Radar chart · Year-wise trends · Stacked risk breakdown · Bob's trend insights

Real-World Results & Scale

30%

Target dropout
reduction

₹12L+

Saved per semester
at CHARUSAT

5 min

Setup time — import
your ERP CSV

4500+

Indian colleges
that need this



CHARUSAT-Ready

Works with CHARUSAT FMIS CSV export today — coordinators can onboard in 5 minutes with zero IT involvement or integration work.



Any University

CSV-based import means EduGuard scales to any Indian university. NEP 2020 mandates outcome tracking — massive national demand.



Future: Live API

IBM Bob MCP + FMIS API enables real-time sync. Risk scores auto-update every time attendance is marked in the ERP system.



National Scale

4,500+ engineering colleges in India face this exact dropout crisis. EduGuard scales to 10,000+ students via Bob subagents.

Why EduGuard Wins

Approach	Limitation ❌	EduGuard ✅
Manual Coordinator Review	❌ Reactive — intervenes after dropout decision is already made	✅ Real-time AI monitoring for every student continuously
Attendance-Only Alerts	❌ Misses financial stress, social issues, backlog signals	✅ 8-factor model captures the full student picture
Generic Counselling	❌ Same advice for every student regardless of situation	✅ IBM Bob Plan Mode: unique action plan per student
ERP Dashboard	❌ Raw data only — no AI, no insight, no priority ranking	✅ AI risk score + ranked actionable interventions
End-of-Semester Review	❌ Always too late — dropout decisions made in Semester 1–2	✅ Flags risk in Semester 1 within the critical window

🔑 IBM Bob is load-bearing, not decorative — removing Bob breaks EduGuard's core functionality. Agent Mode drives scoring, Plan Mode drives interventions, MCP connects to live institutional data.

Ready to Eliminate Dropout Risk

EduGuard AI · Problem E1 · IBM BoB AI Innovation Hackathon 2026 · CHARUSAT



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AI/ML &
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Kevin

UI/UX Design &
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Dharm

Research &
Presentation Strategy



github.com/dhyeyborad7/bob-ai-hackathon-EduGuard



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